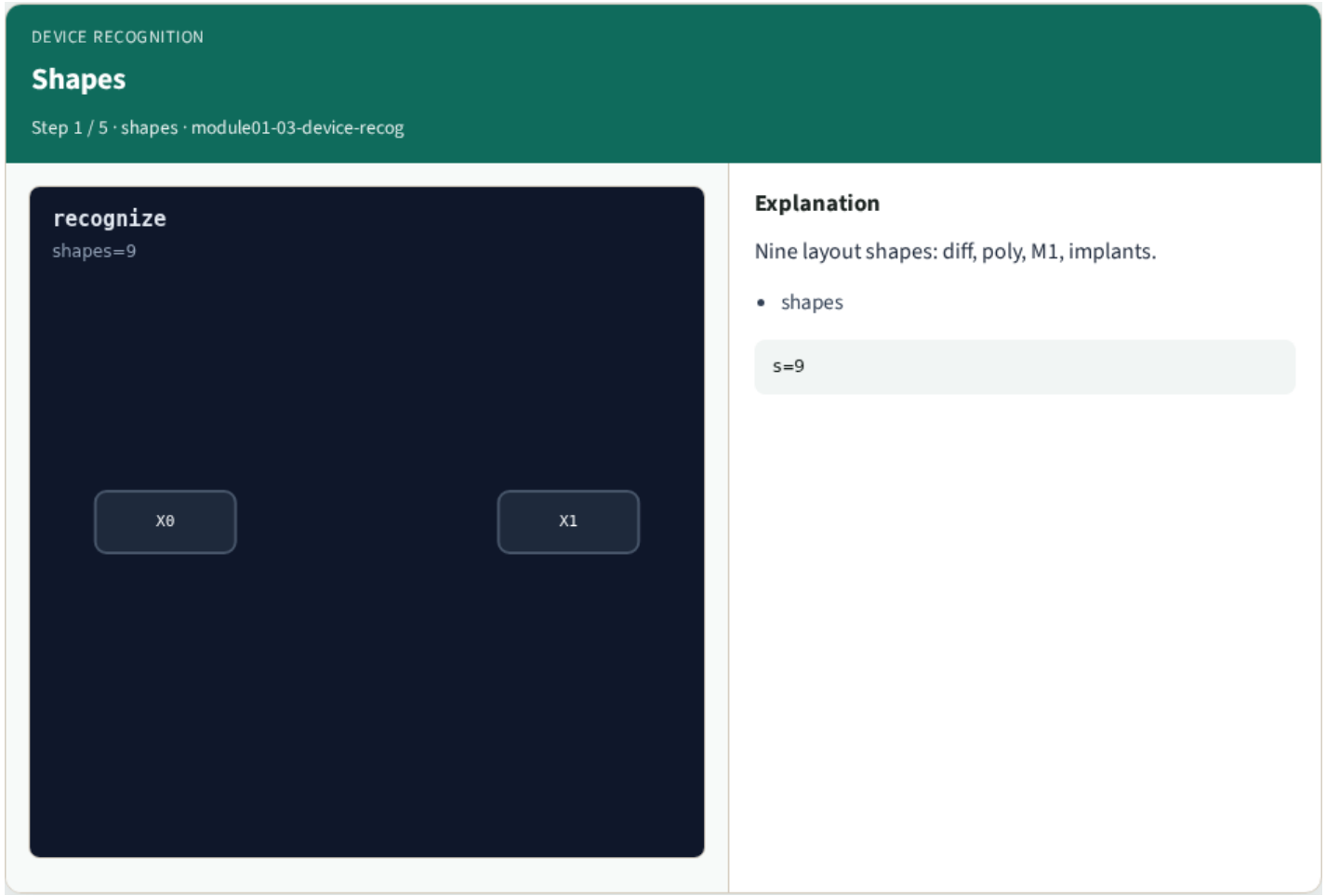
Device recog

Device recognition maps layout shapes to MOS instances , nmos vs pmos

The idea

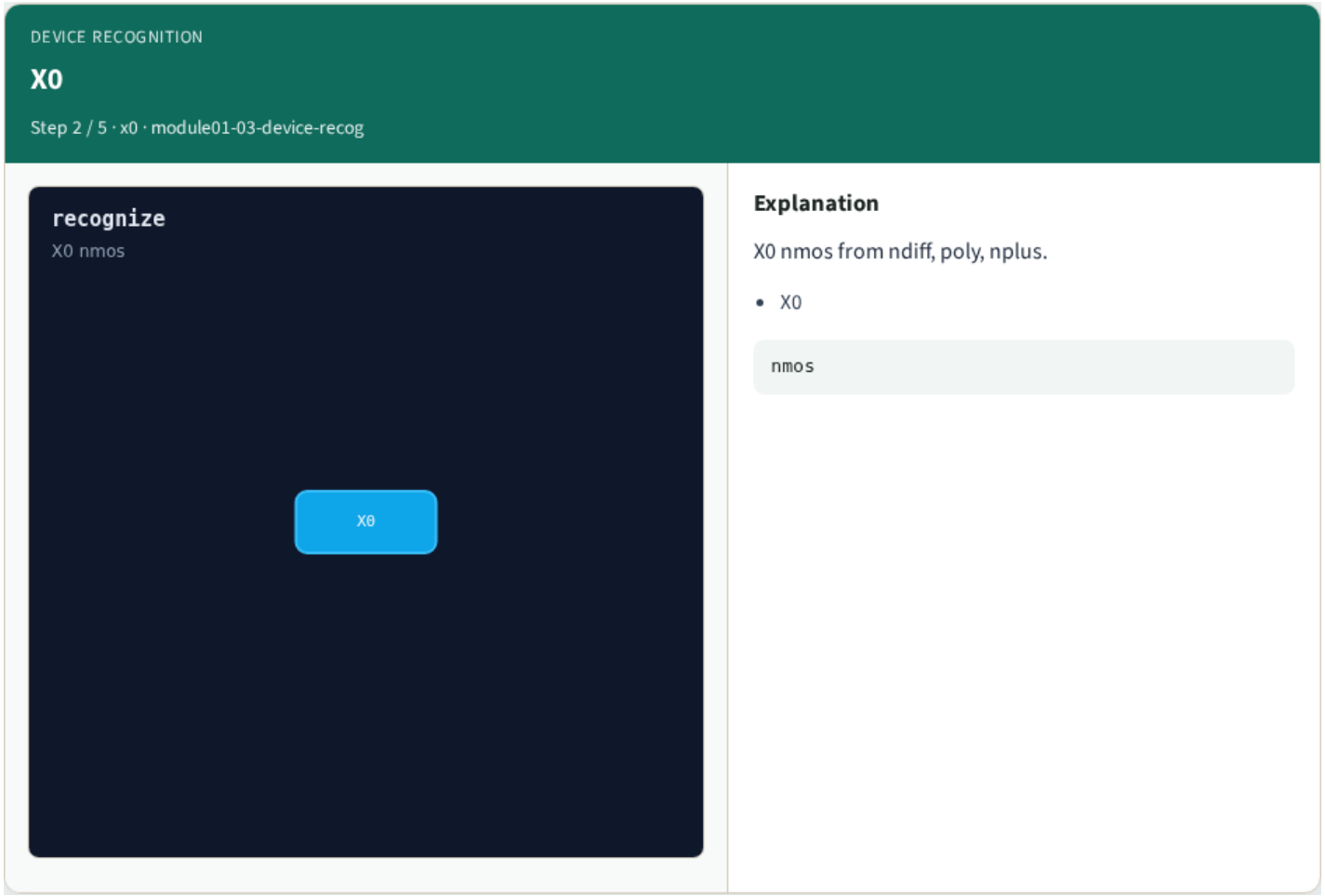
- GOLDEN: nmos=1 pmos=1 (X0/X1)

Shapes



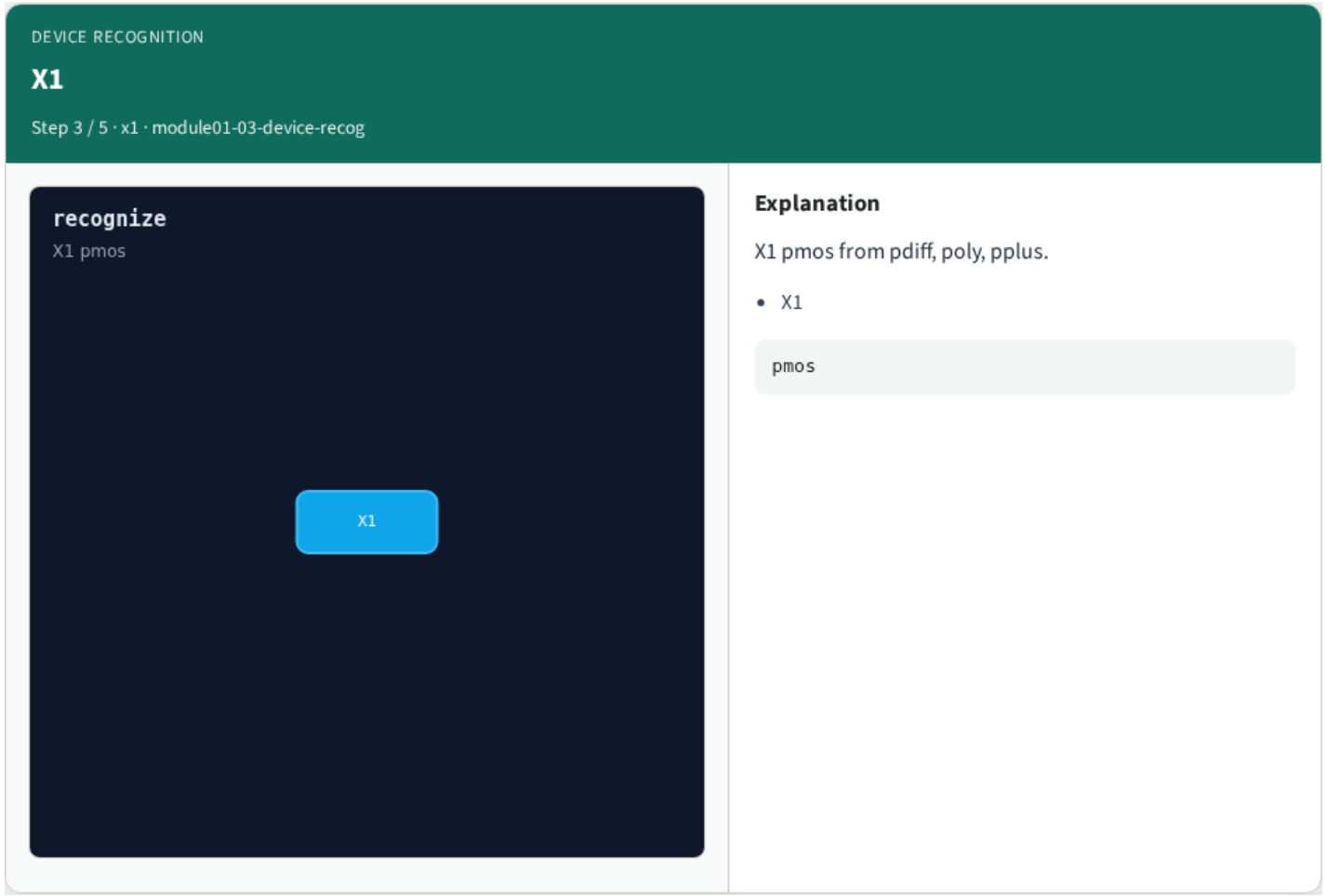
Shapes

X0


DEVICE RECOGNITION
X0
Step 2 / 5 · x0 · module01-03-device-recog
recognize
X0 nmos
X0
Explanation
X0 nmos from ndiff, poly, nplus.
X0
nmos

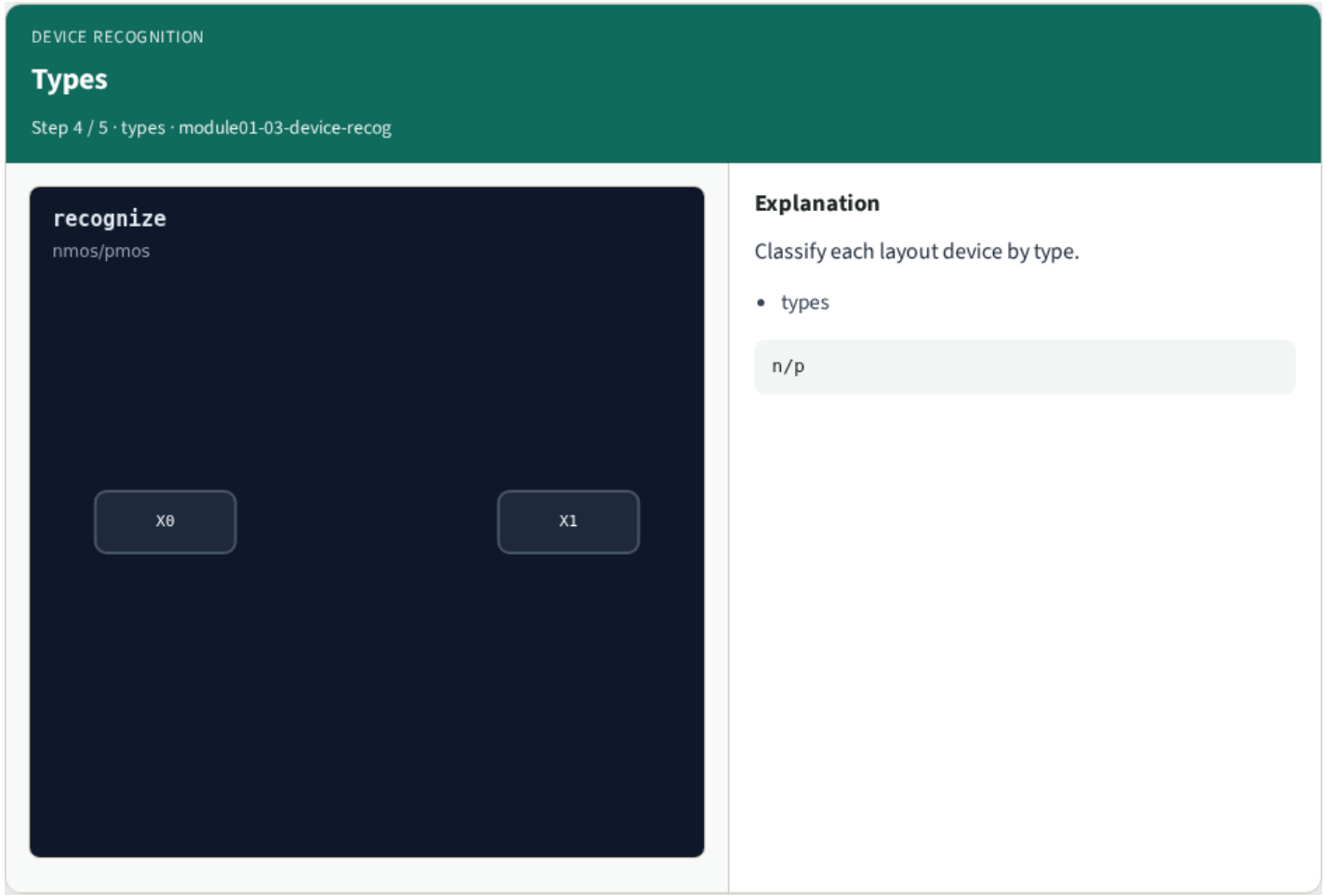
X0

X1


DEVICE RECOGNITION
X1
Step 3 / 5 · x1 · module01-03-device-recog
recognize
X1 pmos
X1
Explanation
X1 pmos from pdiff, poly, pplus.
X1
pmos

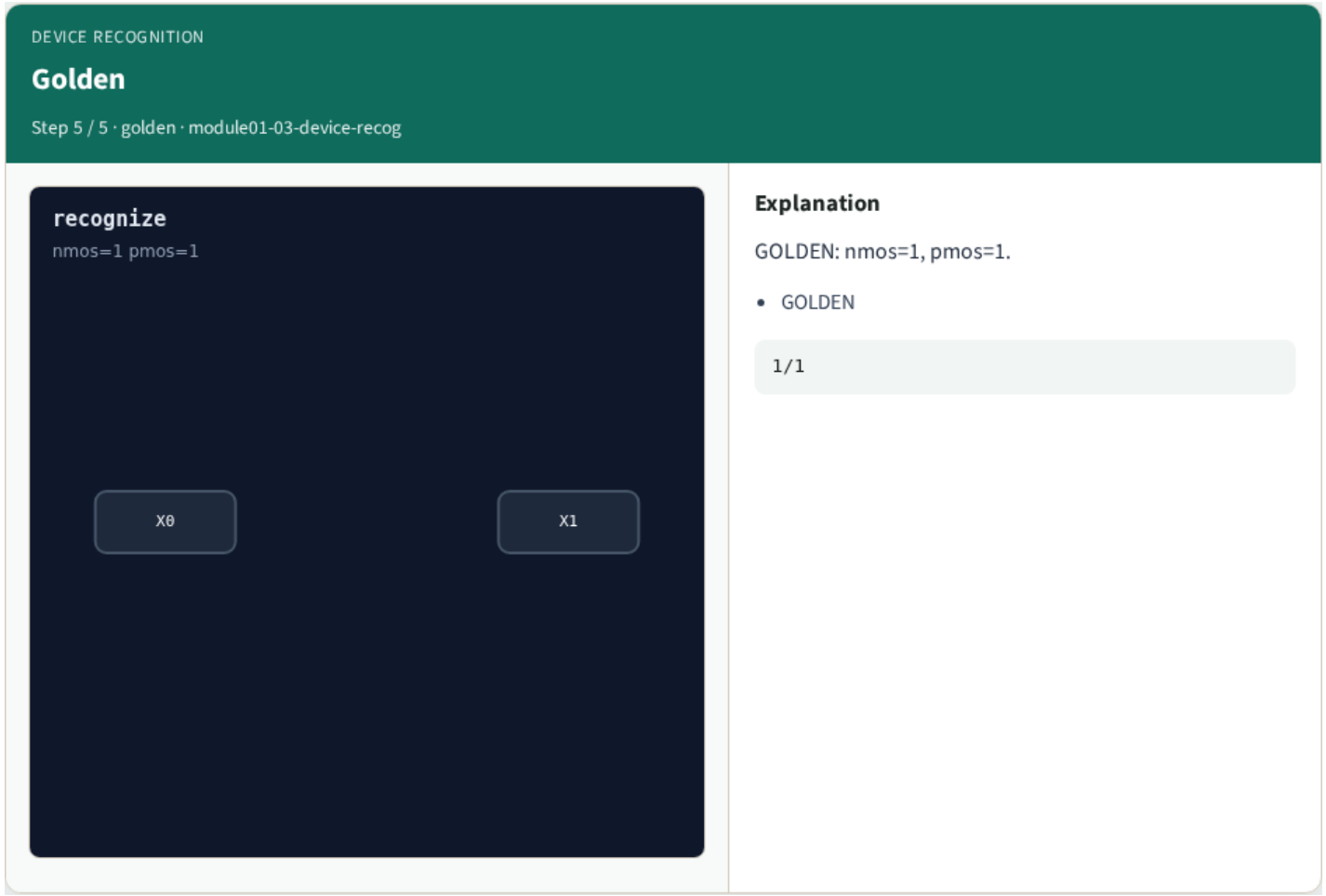
X1

Types


DEVICE RECOGNITION
Types
Step 4 / 5 · types · module01-03-device-recog
recognize
nmos/pmos
X0
X1
Explanation
Classify each layout device by type.
types
n/p

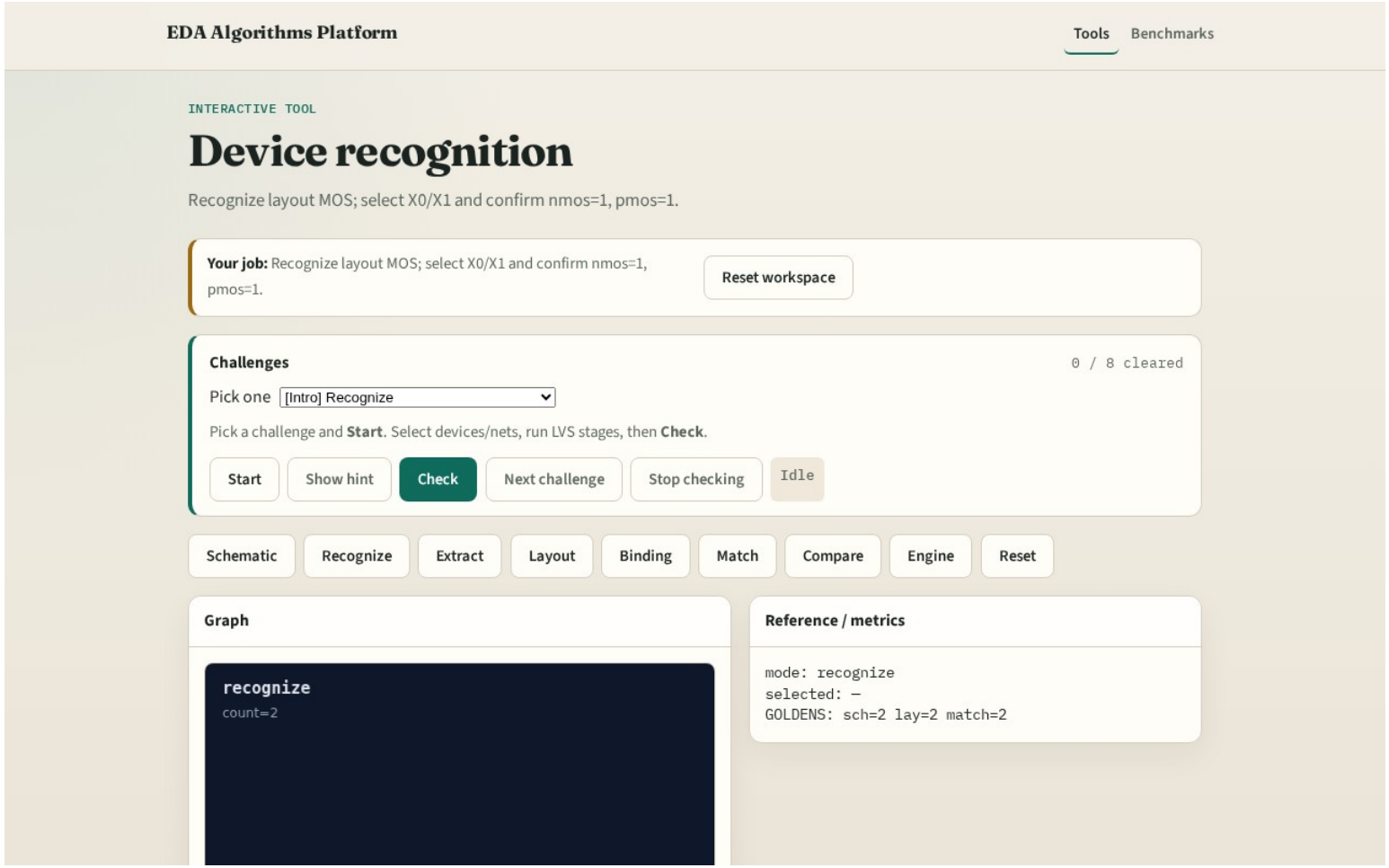
Types

Golden



Golden

Browser lab track



Browser lab starter

Implement track

- Call recognize_devices(); assert nmos==1 and pmos==1

Pitfalls

- Counting shapes instead of devices
- Swapping nmos/pmos polarity

Your turn

- Next: net extraction